SpecEyes: Accelerating Agentic Multimodal LLMs via Speculative Perception and Planning

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Research Questions

- How can we accelerate agentic multimodal large language models that suffer from sequential bottlenecks due to iterative tool invocations?
- Can a lightweight non-agentic model effectively predict when expensive tool-assisted reasoning is unnecessary for visual queries?
- What mechanisms can reliably gate speculative answers without requiring ground-truth labels for confidence estimation?

Core Idea

Agentic multimodal LLMs achieve strong reasoning by iteratively invoking visual tools (zoom, crop, OCR), but this creates a sequential bottleneck where each tool call depends on the previous one, preventing parallelization and causing linear latency growth. SpecEyes addresses this by using a small, tool-free model to speculatively answer queries that don't actually need complex tool chains, while a novel 'answer separability' confidence measure decides when to trust the small model versus falling back to the full agentic pipeline. The key insight is that many queries can be answered from the original image alone, and by identifying these cases early, the system can bypass expensive tool sequences entirely while maintaining accuracy.

Key Formula

Answer Separability Score

$$S_{sep}^{(n)} = \frac{\ell_{[1]}^{(n)} - \mu_K^{(n)}}{\sigma_K^{(n)} + \epsilon}$$

$\ell_{[1]}^{(n)}$ is the top logit for token n, $\mu_K^{(n)}$ and $\sigma_K^{(n)}$ are mean and std of top-K logits, ε is stability constant. Measures how far the leading prediction stands from competitors in a scale-invariant way, avoiding softmax miscalibration issues.

Expected Latency

$$E[L_{SpecEyes}] = c_J + \beta c_S + (1 - \beta\alpha) L_{agent}$$

c_J is judgment cost, c_S is small model cost, β is tool-free screening ratio, α is acceptance rate, L_{agent} is full agentic cost. Shows how speculative bypass reduces expected latency when βα is large.

Key Figure

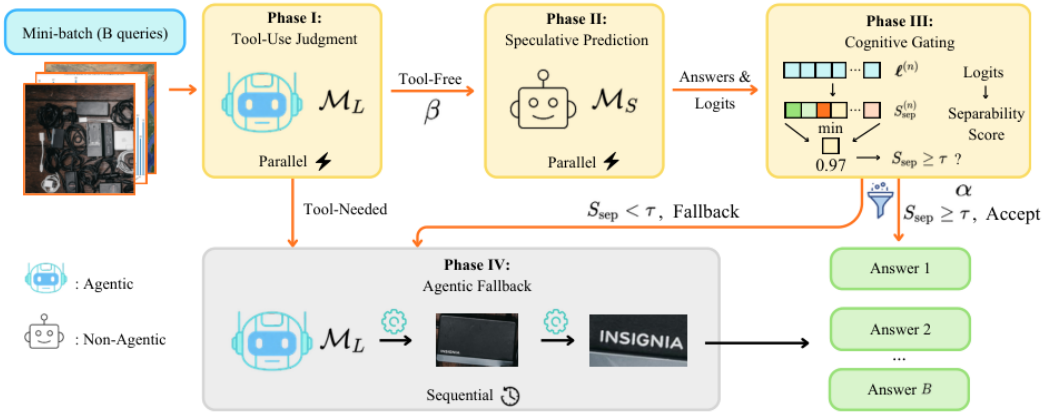

Fig. 2. Pipeline overview of SpecEyes. A batch of B queries passes through a four-phase funnel. **I:** $\mathcal{M}_L$ screens tool necessity, splitting queries into tool-free and tool-required. **II:** A stateless $\mathcal{M}_S$ speculatively answers all tool-free queries with token-level logits. **III:** An answer separability score S_{sep} gates each answer; those above τ are accepted directly. **IV:** Remaining queries fall back to the full agentic loop. The funnel yields $\approx 1/(1-\beta\alpha)\times$ throughput speedup.

3.1 Modeling the Stateful Bottleneck of Agentic MLLMs

Preliminaries. We formalize an agentic multimodal large language model (MLLM) as a stateful reasoning system $\mathcal{A} = (\mathcal{S}, \mathcal{T}, \pi)$, where $\mathcal{S}$ denotes the state space, $\mathcal{T} = \{t_1, \ldots, t_N\}$ is a finite set of perception tools (e.g., `Zoom-in`, `Crop`, `OCR`), and π is policy that jointly selects tool invocations and generates reasoning tokens.

Given a query q and an input image I, the model maintains a state trajectory $\{s_0, s_1, \ldots, s_D\}$ over D reasoning steps. The initial state is $s_0 = (q, I)$. At each step d, the policy produces an action $a_d = \pi(s_d)$ that either invokes a tool $t \in \mathcal{T}$ or emits a final answer. When a tool is invoked, the state transitions as:

$$s_{d+1} = f(s_d, t_d(s_d)), \tag{1}$$

where $t_d(s_d)$ applies the selected tool t_d to the current visual context (e.g., cropping a region of interest from I) and f fuses the resulting observation into the next state. We refer to D as the *agentic depth* of the query.

State Dependency and Sequential Bottleneck. A critical property of Eq. (1) is that subsequent tool selections depend causally on prior observations. Concretely, let $t_{d+1} \sim \pi(\cdot \mid s_{d+1})$ be the tool chosen at step $d{+}1$. Since s_{d+1} contains the output of t_d, the Markov chain $(s_0, a_0, s_1, a_1, \ldots)$ forms a strict data dependency:

$$p(a_{d+1} \mid s_0, a_0, \ldots, s_d) = p(a_{d+1} \mid s_d, t_d(s_d)) \neq p(a_{d+1} \mid s_0). \tag{2}$$

This dependency renders the agentic pipeline inherently *sequential*: step $d{+}1$ cannot begin until step d completes. Consequently, the end-to-end latency for a single query scales linearly with agentic depth:

$$L_{\text{agent}}(q) = \sum_{d=0}^{D(q)} \Big(\underbrace{c_{\text{llm}}}_{\text{reasoning}} + \underbrace{c_{\text{tool}}(t_d)}_{\text{perception}} \Big), \tag{3}$$

where c_{llm} and $c_{\text{tool}}(t_d)$ denote the latency of LLM inference and tool execution at step d, respectively.

Throughput Implication. At the system level, this strict serialization also limits concurrency. Consider a serving scenario with a batch of B queries $\mathcal{Q} = \{q_1, \ldots, q_B\}$. Due to the stateful nature of each query, the large agentic model $\mathcal{A}$ can only process one tool-use loop at a time per query, resulting in a per-query occupancy of $L_{\text{agent}}(q_i)$. The maximum throughput is therefore bounded by:

$$\Theta_{\text{agent}} \leq \frac{B}{\sum_{i=1}^{B} L_{\text{agent}}(q_i)}. \tag{4}$$

Figure 2: Shows the complete four-phase SpecEyes pipeline with parallel processing funnel, illustrating how queries flow through tool-use judgment, speculative prediction, cognitive gating, and agentic fallback.

Limitations

- Currently operates only at agentic depth D=0 (fully tool-free), limiting speedups on benchmarks where most queries genuinely require tool assistance like HR-Bench
- Gating threshold requires calibration on a small held-out validation set, adding deployment complexity
- Performance gains depend heavily on the screening ratio β and acceptance rate α, which vary significantly across different query types and domains
- The small speculative model (Qwen3-VL-2B) may lack capability for complex reasoning tasks that could benefit from shallow tool usage